SOMAIYA VIDYAVIHAR UNIVERSITY

05/05/2025 (E)

Semester:– January 2025 – April 2025 Maximum Marks: 50 Examination: ESE Examination Duration:2 Hrs.		
Programme code: 01 Programme: BTech COMP	Class: TY	Semester: VI(SVU 2020)
Name of the Constituent College/School K. J. Somaiya School of Engineering	Name of the department: Computers	
Course Code: 116U01C601	Name of the Course: Digital Signal and Image Processing	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Question No.	SECTION A	Marks
Q 1.1	In digital signal processing, signals are represented in: a) Analog form b) Discrete-time form c) Continuous-time form d) None of the above	01 M
Q 1.2	If the output of the system of the system at any 'n' depends only the present or the past values of the inputs then the system is said to be _____ a) Linear b) Non-Linear c) Causal d) Non-causal	01 M
Q 1.3	The process of converting an analog signal into a digital signal is called: a) Modulation b) Sampling c) Filtering d) Quantization	01 M
Q 1.4	Which of the following statement is incorrect about DIT- FFT? a) It requires complex additions of 'N log2N.' b) The number of input samples is given by 2^i. c) The input sequence is represented in bit-reversal order. d) The output sequence is represented in bit-reversal order.	01 M
Q 1.5	Define spatial domain filters.	02 M
Q 1.6	DIT algorithm divides the sequence into a) Even and odd samples b) Upper higher and lower spectrum c) Small and large samples d) Positive and negative values e)	01 M

Q 1.7	Time reversal of a discrete time signal refers to a) y[n] = x[-n+k] b) y[n] = x[-n] c) y[n] = x[-n-k] d) y[n] = x[n-k]	01 M
Q 1.8	List different types of LTI systems.	02 M

Que. No.	Question	Max. Marks
Q2	**Solve any Four**	20
i)	Determine whether the discrete-time signal is periodic or non-periodic: $$x(n) = cos\,(\frac{3\pi n}{4})$$	5
ii)	Prove that DFT matrix is symmetric? Justify your answer.	5
iii)	What is the role of a mask (or kernel) in spatial filtering? Explain with an example.	5
iv)	Compute Walsh transform of the given image. 2 1 2 1 1 2 3 2 2 3 4 2 1 2 3 2	5
v)	What effect would set to zero the lower order 2 bits in 4BPP image planes have on the histogram of an image in general?	5
vi)	Explain morphological opening and closing	5

Que. No.	Question	Max. Marks
Q3 A	For a 3 bit 4*4 size image , perform the following operations: a) Negation b) Thresholding with T=4 c) Intensity level slicing with background r1=2 and r2=5 d) Bit plane slicing for MSB and LSB planes e) Clipping with r1=2 and r2=5 1 2 3 0 2 4 6 7 5 2 4 3 3 2 6 1	10
	OR	
Q3 A	Construct the block diagram and signal flow graph of the discrete time system whose input-output relations are described by following difference equation y(n) = 0.6y(n+1)+x(n)-4x(n-8)	10

Q 3 B	Solve	10
i)	An 8 point sequence is given by x (n) = [1, 1, 1, 1, 0, 0, 0, 0]. Compute 8-point DFT of x (n) by radix-2 DIT-FFT.	10